

Facility Layout Optimization For FMCG Production Line

Mustafa Cesmeli

NOVEMBER 2025

Manufactured Products and Product Family Analysis

The enterprise analyzed within the scope of the project operates in the fast moving consumer goods sector and carries out high volume production in the chocolate products, biscuits, and coated confectionery categories. The enterprise manufactures under 2 main brands with high national and international recognition. The common characteristics of the manufactured products are determined as requiring sensitive heat treatment, having a fragile structure, containing high hygiene standards, and requiring the continuous progression of production processes. The production strategy is based on a continuous production model functioning on a 3 shift basis. The production lines located within the facility are grouped according to the physical form of the products and processing technology with a product family logic, classified as product specific and flexible.

The first family, operating with a continuous flow production type, includes dough based baked and chocolate filled coated premium biscuit products. The second family, operating with an assembly oriented production type, consists of products in the form of a chocolate shell containing surprise toy materials inside. The bulk and stand products family, operating with a manual and semi automatic production model, covers the bulk display preparation and bulk packaging operations of products containing toys. The focus product family, which has a hybrid structure where automatic and manual operations are carried out together, includes thin and long baked cracker products coated with chocolate and various particles.

Production Line	Product Family	Product Description and Details	Production Type
Line 1 / Line 2	Bakery Group	Dough-based, baked, and chocolate-filled or coated premium biscuit products.	Continuous Flow
Line 3 / Line 4	Chocolate & Assembly Group	Chocolate shell products containing toy materials inside, targeted at the children's demographic.	Assembly-Oriented
Line 5	Bulk Packaging Group	Bulk packaging and mass display preparation operations for the chocolate and assembly group products.	Manual / Semi-Automatic
Line 6 / Line 7	Coated Stick Group	Thin, long baked cracker products coated with chocolate and particle toppings. (<i>Project Focus</i>)	Hybrid (Automatic + Manual)

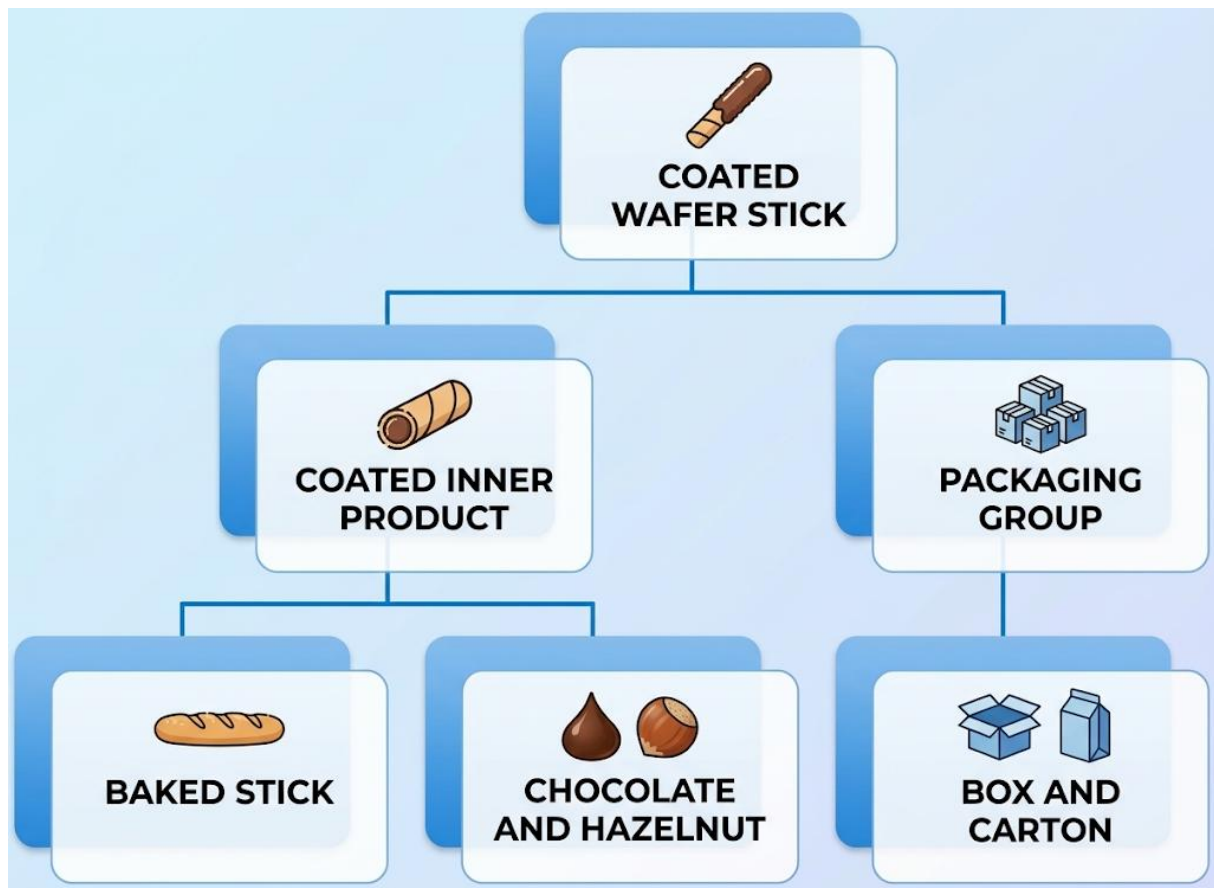
Focus Product Family Analysis

The coated stick cracker product family, which is the main focus point of the facility planning project, has special requirements in terms of logistics and layout. The products manufactured in the relevant production lines are divided into 3 main variations. These are standard stick crackers coated with chocolate, stick crackers coated with chocolate and puffed rice, and stick crackers coated with chocolate and hazelnut particles.

This product group has a high fragility coefficient due to its long and thin structure. After exiting the baking oven, it must be transferred to the cooling and packaging line over the shortest possible distance. The long transportation distances present in the current layout plan increase the risk of product damage in this group. Furthermore, after the products are automatically packaged, the

secondary packaging process conducted in the form of box filling is carried out manually by operators. This situation creates a need for a large operational area at the end of the line.

The product tree consists of 2 main groups. These are the coated internal product and the packaging group. While the coated internal product component consists of the baked stick along with chocolate and nut particles, the packaging group includes the box and carton components.



Focus Product Production Process

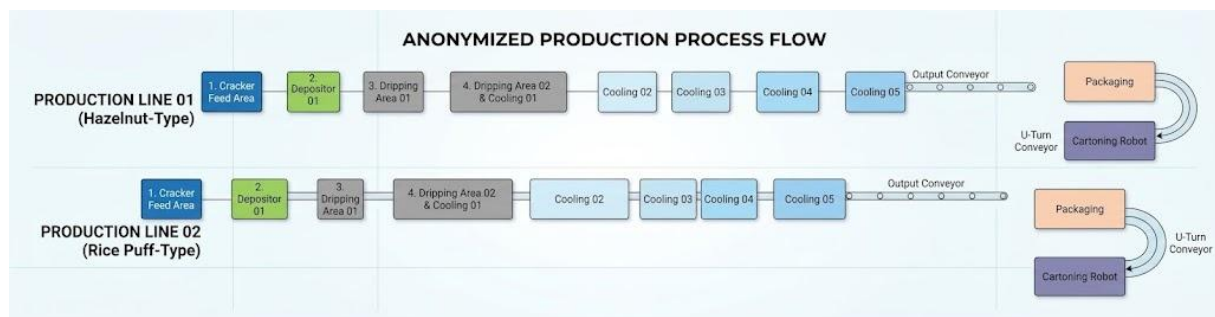
The enterprise maintains continuous production 24 hours a day with a 3 shift system in order to ensure continuous efficiency. The coated stick cracker production process subject to the project is a sequential operation starting from the entry of raw materials into the factory until their transfer to the smart warehouse. The process steps are explained sequentially below.

The process begins with the preparation of the biscuit dough. The prepared dough is shaped and baked in the central ovens located at the head of the facility. The product obtained at the oven exit is a plain and hot semi finished biscuit stick that has not yet been coated with chocolate. In the transfer stage of the process, which contains a physical disconnection, these semi finished sticks exiting the oven are not directly connected to the rest of the production line. The baked sticks are taken from the oven exit, transported over a certain distance within the factory, and manually fed into the starting point of the coating unit.

The plain sticks loaded onto the production line are transferred to the coating station called the depositor. Fluid chocolate sauce is poured over the sticks according to the relevant product recipe. If the manufactured product has a structure with hazelnut or puffed rice, the relevant particles are sprinkled over the chocolate at this stage. The products coated with hot chocolate and particles are transferred directly to the cooling tunnel in order not to distort the current form.

The products advancing on the belt are cooled gradually throughout the tunnel, ensuring the chocolate on them freezes. At the tunnel exit, the product reaches room temperature and becomes ready for packaging.

The cooled products are first converted into single or multiple packages through fully automatic machines without human contact. The packages exiting the machine are poured onto the belt at the end of the line. Operators working at the edge of the belt collect the flowing packages and manually arrange them into shipment boxes. The boxes filled and closed by the operators are placed on pallets. The prepared pallets are transferred to the smart warehouse system constituting the logistics center of the facility and are kept in the stock area until the shipment process occurs.

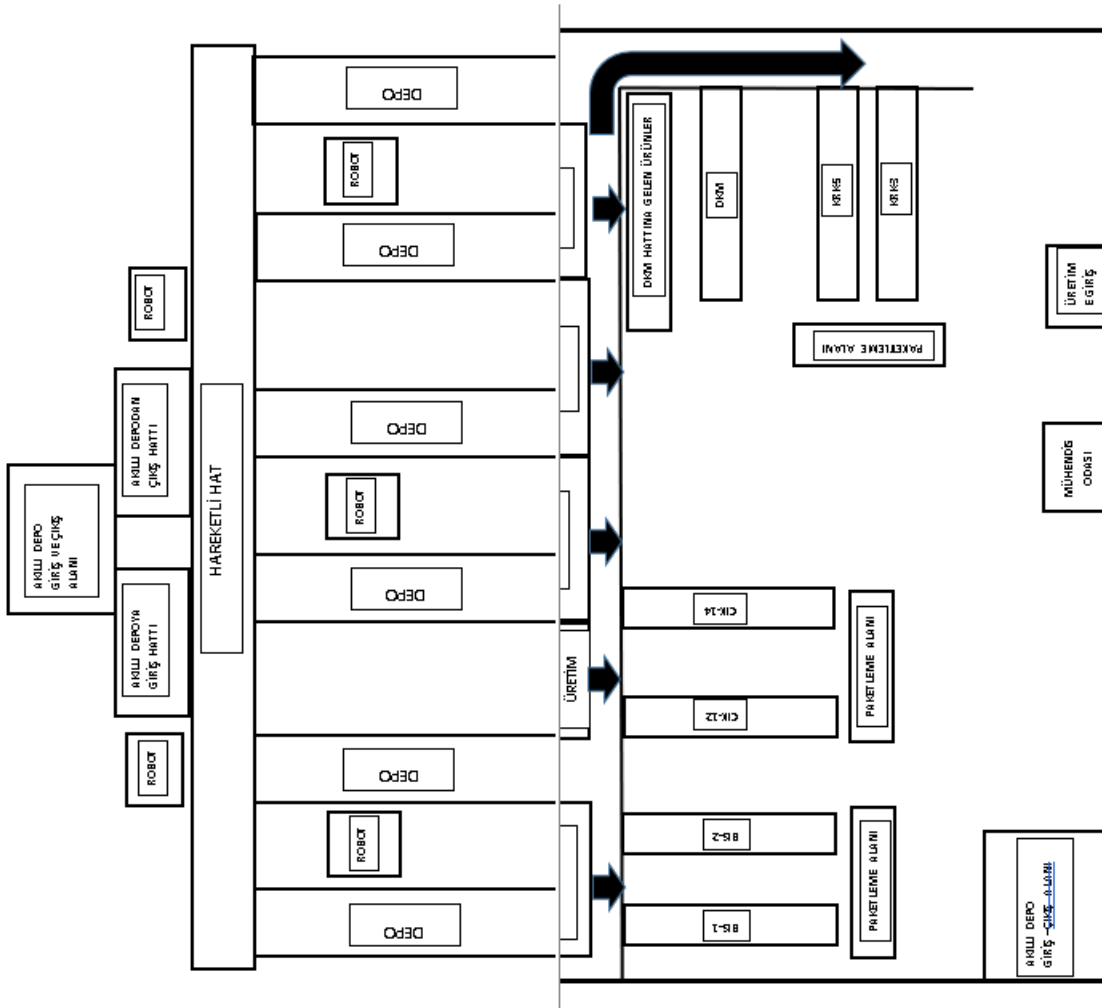
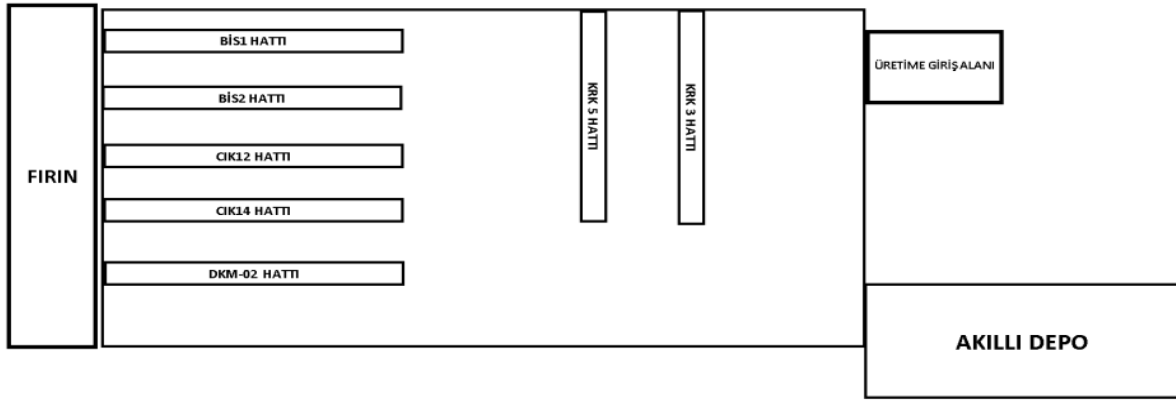


Classification of Product Families and Flow Analysis

The wide product range within the facility was analyzed using the group technology principle by considering production routes, process similarities, and machine requirements. As a result of the examinations, the production structure in the facility was gathered under 3 basic product families. The primary group, the bakery products family, covers cookie and filled biscuit derivatives. The most distinctive characteristic of this family is having a continuous and fully linear flow where the oven exit is directly integrated into the production line. The secondary group, the chocolate and assembly family, includes products with surprise toys. This group exhibits a structure close to cellular manufacturing logic since it involves processes such as assembling parts, molding, and mounting rather than solely food production.

The tertiary group constituting the main focus point of the project was defined as the coated stick family. This family, which includes hazelnut, puffed rice, and plain variations, separates from other families due to structural differences in the production flow. The comparative flow analysis conducted on a family basis clearly reveals the layout efficiency throughout the facility. While raw materials in the bakery products family progress from the oven entrance to the packaged product stage without leaving the line, the process in the coated stick family contains a physical disconnection. The fundamental reason for this disconnection was identified as the distance between the oven exit and the coating unit. The baked sticks in a semi finished state are transferred to the next operation by transport vehicles rather than through a continuous flow.

Furthermore, the flow analysis of the relevant family highlighted an operational difference at the end of the line. While packaging processes in other families are carried out with full automation, the secondary packaging process in the focus product family is completed via a human labor based manual boxing operation. As a result of the analyses, it was determined that the bakery products group is in an ideal state in terms of layout, whereas it was proven that the coated stick group requires layout optimization due to both the transfer distance between the oven and the line and the manual operation area requirement at the end of the line.



Digitization of Flow and Activity Relationships

The production process was modeled based on the assumption that semi finished and finished products are transported on a box basis. The production amount was accepted as 1200 boxes daily considering the continuous production structure with 3 shifts. The flow possesses a linear structure and the operations take place among the oven, transfer area, coating unit, cooling tunnel, automatic packaging, manual boxing, and smart warehouse entry point. The relevant stations are ordered sequentially and flow volumes are defined linearly on the matrix as 1200 units daily.

Code	Department / Activity	From \ To	A	B	C	D	E	F	G
A	Oven (Baked Goods Production)	A	–	1200	0	0	0	0	0
B	Transfer Area (Transportation + Feeding)	B	0	–	1200	0	0	0	0
C	Coating / Depositor	C	0	0	–	1200	0	0	0
D	Cooling Tunnel	D	0	0	0	–	1200	0	0
E	Automatic Packaging (Flowpack)	E	0	0	0	0	–	1200	0
F	Manual Cartoning	F	0	0	0	0	0	–	1200
G	Smart Warehouse Entry	G	0	0	0	0	0	0	–

Quantitative flow data were supported by qualitative proximity requirements between activities. Proximity degrees were assigned by taking flow continuity, product fragility, hygiene requirements, and manual operation density into consideration. The relationships between the oven exit and the transfer area, the transfer area and the coating unit, the coating unit and the cooling tunnel, and the cooling tunnel and automatic packaging have an absolutely necessary proximity degree. The relationships between automatic packaging and the manual boxing area, and between the manual boxing area and the smart warehouse entry are classified with an especially important proximity degree. It was determined that the distance between the oven and the coating unit, which forms the basis of the problem, must be minimized, and in this context, an especially important proximity degree was assigned. An unimportant proximity degree was defined among other stations that harbor an indirect relationship.

Pair	Proximity	Reason
A–B	A	Oven exit is the source of “discontinuity”; semi-finished product transfer is
B–C	A	Transfer is feeding to coating; risk of delay/scrap
C–D	A	Immediate cooling is required after coating; form must not be distorted
D–E	A	Direct packaging flow after cooling
E–F	E	Package exit reverts to manual cartoning; end-of-line labor is intensive
F–G	E	Cartons to pallet → sent to warehouse; logistic proximity is important
A–C	E	Main problem is “oven-coating distance”; it should be as close as possible
A–D	I	Proximity in the flow chain is good, but it is a priority
A–E	O	No direct contact
A–F	U	No direct relationship
A–G	U	No direct relationship
B–D	I	After transfer, the product going to cooling passes through coating; indirect
B–E	O	Indirect
B–F	U	Indirect
B–G	U	Indirect
C–E	I	Packaging after cooling;
C–F	O	Indirect
C–G	U	Indirect
D–F	O	Manual operation after packaging; indirect
D–G	U	Indirect
E–G	I	Warehouse is the goal in the packaged product flow; but F intervenes

Area Requirements

The area requirements of the stations were calculated as the sum of the equipment area, operation area, buffer stock, and service areas. According to the area needs determined in square meters, the oven exit requires 150, the transfer area 50, the coating unit 50, the cooling tunnel 200, the automatic packaging 100, the manual boxing 150, and the smart warehouse shipment interface 100 area units. The total layout area was determined as 800 square meters.

Code	Activity	Area Requirement (m ²)
A	Oven exit / semi-finished	150
B	Transfer & feeding area	50
C	Coating	50
D	Cooling tunnel	200
E	Automatic packaging	100
F	Manual boxing	150
G	Warehouse dispatch interface	100
	TOTAL	800

Construction Based Initial Block Layout

An initial block layout was generated via a construction based algorithm utilizing the activity relationship chart and area requirements. The layout units were dimensioned on a grid structure consisting of 4 rows and 8 columns. In this developed arrangement, absolutely necessary proximity targets were achieved, and sequential activities were positioned adjacent to each other. The critical distance between the oven and the coating unit was reduced, thereby satisfying the especially important proximity target. This developed layout was accepted as a valid initial solution that maximizes qualitative proximity targets.

Row	Cücre Seqence
1	A A A B B C C G
2	A A A E E C C G
3	D D D E E F F G
4	D D D F F F F G

Improvement Based Layout Analysis

An improvement based layout algorithm was applied to the system in order to test whether the initial layout obtained by the construction based algorithm could be improved locally. The fundamental objective of the relevant algorithm is to minimize the total material handling cost by performing area exchanges between departments over a given layout arrangement. Distances between departments were calculated using the rectilinear distance method, and unit transport costs were assumed equal for all departments.

The total transportation cost for the initial layout was calculated as 21600 units. Within the scope of the algorithm, all valid pairwise exchanges fulfilling the area equality condition were tested. In the first trial, the transfer area and the coating unit were exchanged, and the cost increased by 22 percent, reaching 26400 units. In the second trial, automatic packaging and the warehouse shipment interface were exchanged, disrupting the relationship between packaging and manual boxing, causing the cost to increase by 19 percent, reaching 25800 units. In the third trial, the oven exit and the manual boxing area were exchanged, resulting in a value of 21600 units, which is identical to the initial cost.

As a result of the conducted analyses, a lower total cost value could not be found. The fundamental reason for this is that the construction based algorithm had already satisfied all critical proximities while generating the initial layout. Although the improvement algorithm could not produce a new and superior solution, it proved that the initial layout obtained via the construction based method possesses optimal characteristics. The current facility plan was examined to determine the distance values between activity centers for the relevant line, and the total transportation cost of the current state was calculated as 28800 units.

General Evaluation and Conclusion

Within the scope of the conducted study, the facility layout problem belonging to the coated stick cracker product family in the fast moving consumer goods facility was examined. The analysis process consisted of the stages of flow digitization, determination of activity relationships, definition of area requirements, and algorithmic layout generation.

The initial layout generated using the construction based algorithm provided a 25 percent reduction in total transportation cost compared to the current facility arrangement. All iterations performed via the improvement based algorithm demonstrated that the obtained solution could not be advanced to a further level. This situation reveals that the presented proposal layout plan is optimal in a local and practical context. Consequently, the developed facility layout model presents an applicable solution that maximizes flow continuity, minimizes the transportation risk of products with fragile structures, and integrates processes requiring manual operations into the production line with high efficiency.

Layout	Total Transport Cost
Current facility	28 800
Initial layout	21 600
Post-CRAFT	21 600